Stochastic Attention via Langevin Dynamics on the Modern Hopfield Energy

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Motivation

Attention heads are deterministic: the same query always returns the same softmax-weighted memory average. Yet many tasks require *generating* novel, plausible patterns from structured memory. Score-based diffusion models solve this for large datasets by learning a neural score function, but they require thousands of training examples and expensive optimization. When data is scarce, there is nothing to train on. We've explored a different approach: instead of learning a score, we can **derive** the exact score function from the modern Hopfield energy landscape. We observe that the gradient of the **modern Hopfield energy** is exactly the identity minus the attention map. Applying **Langevin dynamics** to this energy yields *stochastic attention*: a **training-free sampler** controlled by a single inverse temperature β (Figure 1).

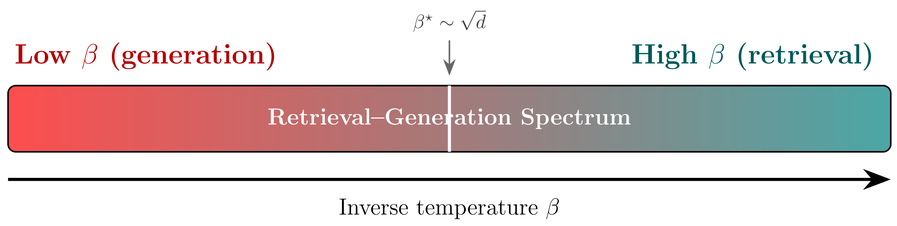


Figure 1. The retrieval–generation spectrum. A single inverse temperature β controls sample behavior: low β yields broad generation, high β yields exact retrieval. The critical temperature $\beta^* \sim \sqrt{d}$ marks the boundary, identified analytically via an entropy inflection condition for random unit-norm memories. A single parameter β interpolates continuously between open-ended generation ($\beta \to 0$) and exact retrieval ($\beta \to \infty$).

Method: Stochastic Attention Update

Suppose you have a memory (pattern) matrix $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ that stores K patterns where each each pattern is given by $\mathbf{m}_k \in \mathbb{R}^d$. To generate a new pattern or retrieve an existing one, we sample from $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$. The Unadjusted Langevin Algorithm (ULA) turns any smooth energy U into a sampler:

$$\boldsymbol{\xi}_{t+1} = \boldsymbol{\xi}_t - \tilde{\alpha}\, \nabla U(\boldsymbol{\xi}_t) + \sqrt{2\tilde{\alpha}}\, \boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

The gradient of the modern Hopfield energy is exactly the attention residual:

$$\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \underbrace{\mathbf{X}\,\mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi})}_{\mathbf{T}(\boldsymbol{\xi}) \,=\, \text{softmax attention map}}.$$

Substituting ∇E into the ULA (with $U = \beta E$, $\tilde{\alpha} = \alpha/\beta$):

$$\boldsymbol{\xi}_{t+1} = \underbrace{(1-\alpha)\,\boldsymbol{\xi}_t}_{\text{contraction}} + \underbrace{\alpha\, \mathbf{X}\,\mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi}_t)}_{\text{attention pull to memories}} + \underbrace{\sqrt{\tfrac{2\alpha}{\beta}}\, \boldsymbol{\epsilon}_t}_{\text{Gaussian noise}}$$

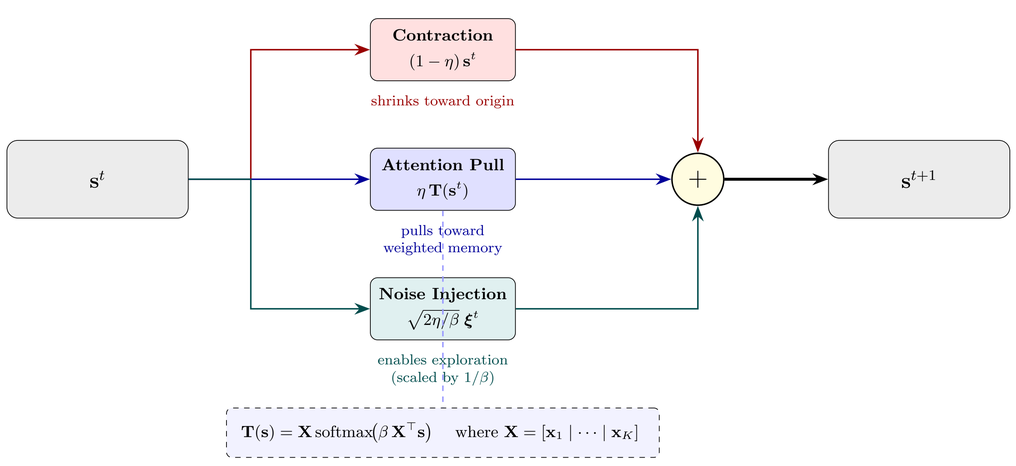


Figure 2. The stochastic attention update decomposes into three components: contraction toward the origin, an attention pull toward stored memories, and isotropic Gaussian noise scaled by temperature. Each step costs $\mathcal{O}(dK)$, with no training required. Convergence is W_2-geometric in the convex regime ($\beta\sigma_{\max}^2 < 2$); Metropolis-Adjusted Langevin Algorithm (MALA) acceptance = 99.2% at $\alpha = 0.01$ confirms negligible discretization bias. The critical temperature β^* is selected via the entropy inflection condition and scales as $\beta^* \sim \sqrt{d}$.

Experiment 1: Phase Behavior & Convergence (Synthetic)

The theory predicts that β acts as a continuous order parameter, with a smooth transition between generation and retrieval centered near $\beta^* \sim \sqrt{d}$. We tested this on synthetic random unit-norm patterns and verified convergence to the correct Boltzmann target (Figure 3).

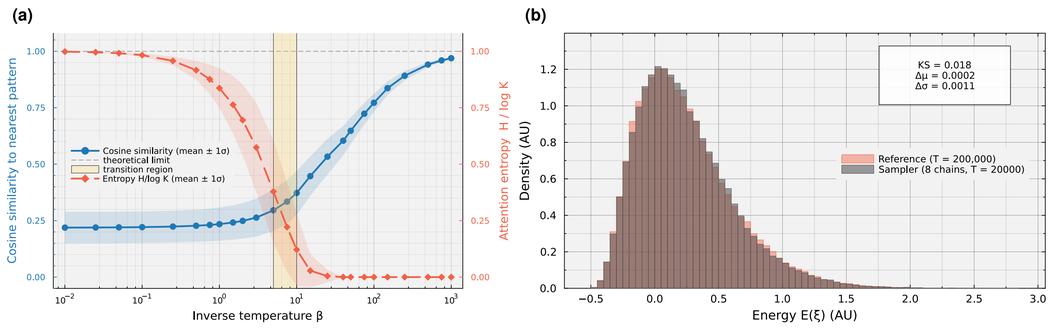


Figure 3. Synthetic data experiments. **(a)** $d = 64$, $K = 16$. Both cosine similarity to the nearest pattern (blue) and attention entropy (coral) underwent a smooth sigmoidal transition centered at $\beta \approx 5$–10 (gold band), consistent with the predicted $\beta^* \sim \sqrt{d} \approx 8$. **(b)** $d = 8$, $K = 4$, $\beta = 5$. Eight independent chains (gray) converged to a long-run reference distribution (coral); Kolmogorov–Smirnov (KS) testing confirmed the sampler reached the correct Boltzmann target.

References

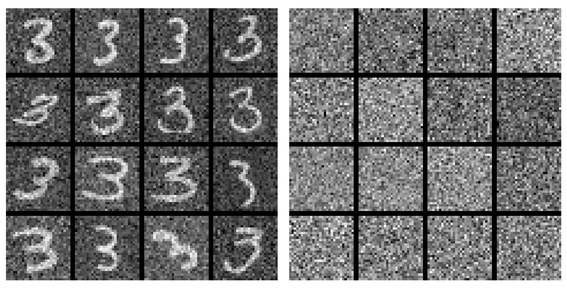
[1] Ramsauer et al., "Hopfield Networks is All You Need," *ICLR*, 2021.
[2] Roberts & Tweedie, "Exponential Convergence of Langevin Distributions and Their Discrete Approximations," *Bernoulli*, 2(4):341–363, 1996.
[3] Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models," *NeurIPS*, 33:6840–6851, 2020.
[4] Durmus & Moulines, "Nonasymptotic Convergence Analysis for the Unadjusted Langevin Algorithm," *Ann. Appl. Probab.*, 27(3):1551–1587, 2017.

Experiment 2: MNIST Image Generation ($K = 100$, $d = 784$)

We stored $K = 100$ images of digit "3" and compared stochastic attention against both synthetic baselines (bootstrap, Gaussian perturbation, convex combinations, Gaussian Mixture Model with Principal Component Analysis (GMM-PCA)) and learned generative models (Variational Autoencoder (VAE), Denoising Diffusion Probabilistic Model (DDPM)). We ran 30 independent chains at $\beta = 2000$ (retrieval) and $\beta = 200$ (generation, near β^*), collecting 150 samples per regime (Figure 4).

Method	Nov.↑	Div.↑	Energy↓
Bootstrap	0.000	0.459	−0.500
Gaussian perturb.	0.004	0.450	−0.496
Convex combo	0.092	0.008	−0.399
GMM-PCA	0.198	0.419	−0.303
VAE (latent = 8)	0.214	0.441	−0.286
DDPM†	0.938	0.991	+0.44
MALA (β=2000)	0.151	0.598	−0.305
SA (β=2000)	0.152	0.600	−0.303
SA (β=200)	**0.548**	**0.885**	+1.467

†DDPM generated isotropic noise.



SA (β=2000) **DDPM**

Figure 4. (Left) Novelty, diversity, and energy scores for all methods on MNIST digit "3" (K=100, d=784). SA at generation temperature (β=200) achieved the highest novelty and diversity among energy-based methods. **(Right)** Sample grids from SA (β=2000, retrieval regime) and DDPM (5,000 training epochs). DDPM produced isotropic noise on this small dataset. In the generation regime, SA achieved **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while matching the Metropolis-corrected MALA gold standard. DDPM failed, producing isotropic noise despite 5,000 training epochs.

Experiment 3: Sequence Generation (Pfam RRM, $K = 68$, $d = 59$)

Protein sequences posed a harder test than images: generated samples must preserve global amino acid composition, position-specific conservation and pairwise co-evolutionary couplings that define family membership. We stored 68 aligned RRM (RNA Recognition Motif) sequences from Pfam, one-hot encoded and PCA-reduced to d=59 (95% variance retained). The entropy inflection gave β^*=3.85, half the random-pattern prediction $\sqrt{d}$=7.68, because conserved residues concentrate the similarity spectrum (Figure 5).

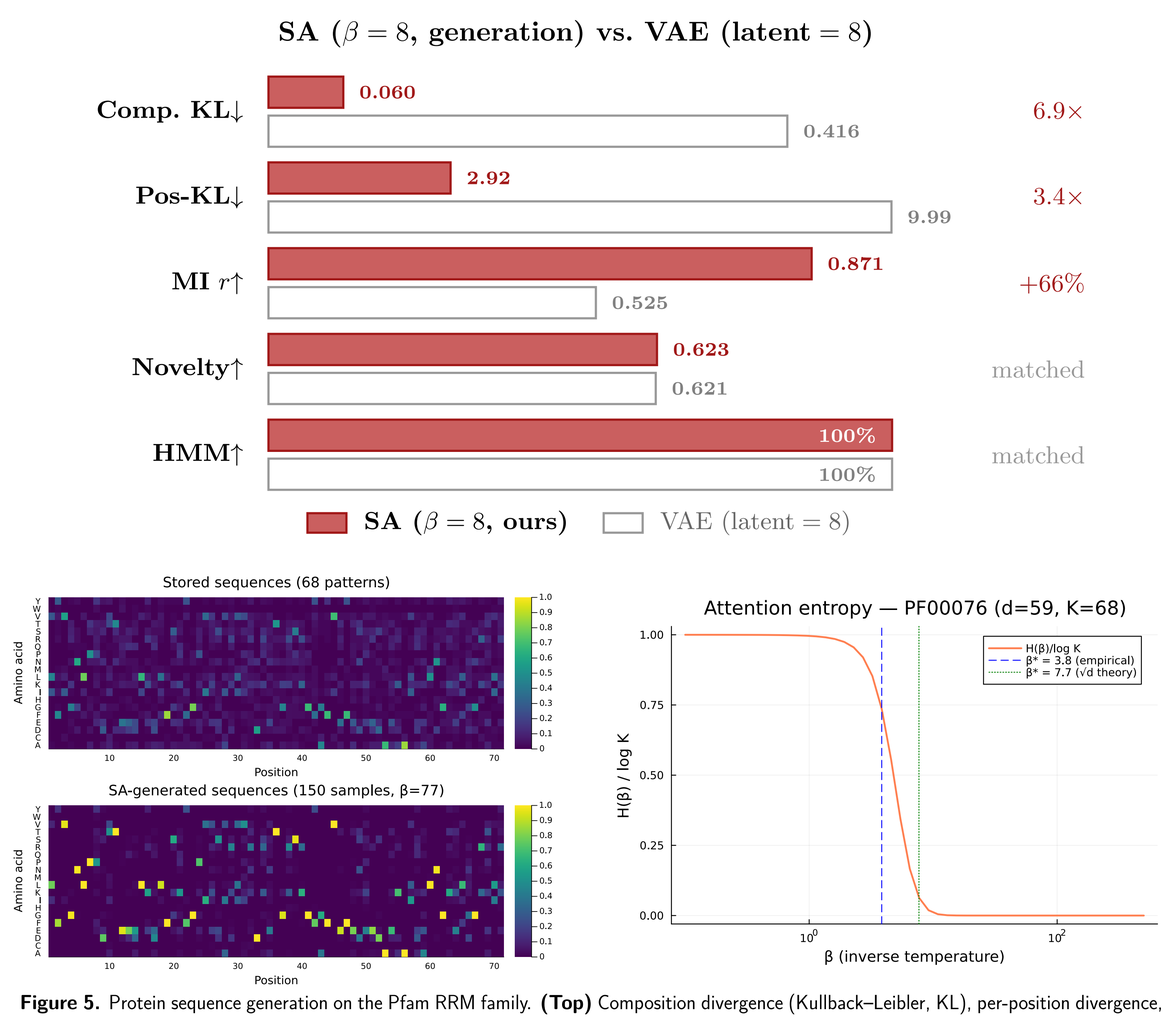


Figure 5. Protein sequence generation on the Pfam RRM family. **(Top)** Composition divergence (Kullback–Leibler, KL), per-position divergence, and coupling fidelity for SA vs. VAE. **(Bottom-left)** Stored (top) vs. SA-generated (bottom) amino acid frequencies. **(Bottom-right)** Entropy inflection identified β^*=3.85, the retrieval–generation boundary.

At generation temperature, SA matched the VAE on novelty while achieving **6.9**× lower composition divergence, **3.4**× lower per-position divergence, and **66**% higher coupling fidelity. Every generated sequence passed the Pfam Hidden Markov Model (HMM) filter (median E-value 8.1×10^{-34}). Because the Langevin gradient was recomputed from stored sequences at every step, it continuously steered samples toward the family's structural profile, fidelity that the VAE, trained on only 68 examples, could not maintain.

Key Result

Stochastic attention is a **training-free** generative sampler whose score function is the attention map itself. Each step costs $\mathcal{O}(dK)$, the same as one attention head, with no learned parameters, no training loop, and no dataset size requirement. A single analytically identified parameter β^* controls the generation–retrieval tradeoff.

On **MNIST** (K=100), SA achieved **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while DDPM failed. On **protein sequences** (K=68), SA achieved **6.9**× lower composition divergence and **66**% higher coupling fidelity than the VAE, with 100% Pfam HMM pass rate.